

The Riemann Hypothesis Up to Height 3000: Exact Counting of Critical-Line Zeros, GUE Statistics, and a Learning Probe

What numerical verification establishes, what it cannot, and a cautionary lesson about sampling

PINNeAPLe Labs

PINNeAPLe Labs — Physics AI research notes

September 25, 2026 · Experiment 06 · Millennium problems · Riemann

Abstract

The Riemann hypothesis (RH) states that every non-trivial zero of $\zeta(s)$ lies on the critical line $\text{Re } s = \frac{1}{2}$. Numerical verification cannot prove it, but it can certify it up to a height T by comparing the number of sign changes of Hardy's Z -function on the line with the total number of zeros $N(T)$ in the critical strip. Up to $T = 3000$ a first scan found 2,467 line zeros against $N(T) = 2,469$. Localising the deficit window by window showed that the two 'missing' zeros are a close pair ($\gamma \approx 1977.1739$ and 1977.2714 , gap 0.0975) that the coarse grid stepped over; after an eight-fold refinement the counts agree ($2,469 = N(T)$), so all zeros with $0 < \text{Im } s < 3000$ lie on the critical line — a tiny, known range (verification exceeds 10^{13} zeros), reproduced independently. The unfolded spacings follow random-matrix (GUE) statistics (Kolmogorov–Smirnov distance 0.040 to the Wigner surmise vs 0.315 to Poisson). A PINNeAPLe GRU forecasts the next spacing from the previous 32 with $R^2 = 0.85$ on a held-out block; the same pipeline gives only $R^2 = 0.21$ on GUE eigenvalues (and -0.08 on a Poisson process), and a linear AR model reaches 0.27 on ζ — about the GUE level. The extra nonlinear predictability of low-lying ζ spacings is therefore NOT explained by GUE rigidity; we report it as an unexplained observation, plausibly a low-height, non-universal effect (the spacing variance, 0.149, is also below the GUE value), and make no claim from it.

Keywords: Riemann hypothesis; zeta zeros; Turing method; random matrix theory; GUE; Montgomery pair correlation; Millennium Prize problems

What this study shows — and what it does not

- Shows: every zero of ζ with $0 < \text{Im } s < 3000$ is on the critical line (count certified against $N(T)$).
- Shows: a sampling-only search silently misses close pairs; comparing with $N(T)$ is what makes the verification trustworthy.
- Does not show RH: finitely many zeros say nothing about the infinitely many beyond; statistics and learning are consistent with RH, not evidence of it.

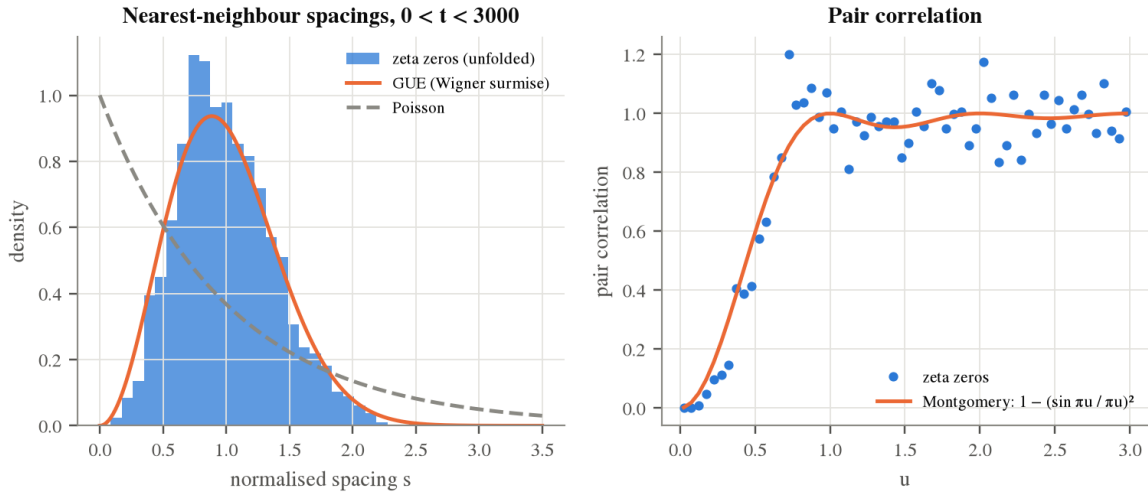
1 Method

- $Z(t) = e^{i\theta(t)} \zeta(\frac{1}{2} + it)$ is real, $|Z(t)| = |\zeta(\frac{1}{2} + it)|$; each sign change brackets a zero on the line (mpmath *siegelz*, 15-digit precision). Grid step = mean spacing/6, each bracket refined by the Illinois method.
- $N(T)$ — the number of zeros with $0 < \text{Im } s < T$ anywhere in the critical strip — from mpmath *nzeros* (argument-principle/Turing-type counting).
- Where the counts differ, the window of width 100 is resampled 8×, 32×, 128× finer until the counts agree.
- Unfolding with the smooth part of $N(t)$: $u = \theta(t)/\pi + 1$; spacing, pair-correlation and KS statistics on the unfolded zeros.
- Learning probe: GRU (PINNeAPLe recurrent family) and linear AR(32) predict the next unfolded spacing from the previous 32; chronological 70/15/15 split; controls: the identical pipeline on shuffled spacings, on unfolded GUE eigenvalues ($N = 3000$) and on a Poisson process. (The probe was run on the first-scan series of 2,467 zeros; the two missing zeros affect a handful of the 2,434 samples.)

2 Results

Table 1. Verification up to T .

Quantity	Value
height T	3000
zeros on the line, coarse scan	2,467
$N(T)$, all zeros in the strip	2,469
windows needing refinement	1
zeros on the line after refinement	2,469
all zeros on the line up to T	True
first zero (check vs <code>mpmath.zetazero(1)</code>)	14.134725142 / 14.134725142
scan CPU time	12.5 h

**Figure 1.** Left: spacing distribution against the GUE Wigner surmise and Poisson. Right: pair correlation against Montgomery's conjecture.**Table 2.** Next-spacing predictability on held-out blocks (R^2 relative to the training mean). A shuffled-spacing control gives $R^2 \approx 0$ for both models, ruling out leakage in the pipeline.

Series	R^2 linear AR(32)	R^2 GRU (PINNeAPPLE)	lag-1 autocorrelation
zeta zeros (unfolded)	0.270	0.848	-0.386
GUE eigenvalues (unfolded)	0.199	0.207	—
Poisson process	-0.023	-0.080	—

3 Discussion

The verification part is standard and exact up to T ; its value here is methodological: sampling-based zero finding is not a certificate, and the comparison with $N(T)$ is indispensable (Turing's insight). The statistical part reproduces the Montgomery–Odlyzko law at low height. The learning part is the kind of result that is easy to over-interpret. A shuffled control rules out leakage; the GUE control shows that random-matrix rigidity alone gives much lower predictability than the GRU finds for ζ at these heights. Candidate explanations we have not tested include the regularity of low-lying zeros relative to Gram points and lower-order (arithmetic) corrections to GUE statistics, both of which fade with height. The decisive test is to repeat the probe on zeros at large height (e.g. Odlyzko's published zeros near 10^{20}), where GUE statistics hold much more closely; until then no conclusion is drawn.

References

- [1] Bombieri, E. The Riemann hypothesis. Clay Mathematics Institute Millennium Problem description (2000).
- [2] Turing, A. M. (1953). Some calculations of the Riemann zeta-function. *Proc. London Math. Soc.* (3) 3, 99–117.
- [3] Montgomery, H. L. (1973). The pair correlation of zeros of the zeta function. *Proc. Symp. Pure Math.* 24, 181–193.

- [4] Odlyzko, A. M. (1987). On the distribution of spacings between zeros of the zeta function. *Mathematics of Computation* 48, 273–308.
- [5] Lehmer, D. H. (1956). On the roots of the Riemann zeta-function. *Acta Mathematica* 95, 291–298.
- [6] Platt, D., Trudgian, T. (2021). The Riemann hypothesis is true up to $3 \cdot 10^{12}$. *Bulletin of the London Mathematical Society* 53, 792–797.
- [7] Johansson, F., et al. mpmath: a Python library for arbitrary-precision floating-point arithmetic, <https://mpmath.org>